

Mursalin Habib

PhD Student · Rutgers University

✉ mursalin.habib@rutgers.edu | 🏠 mh7111997.github.io | 📄 | 🐙

Research Interests

I am broadly interested in all things theoretical computer science. My current research is focused on the theory of error-correcting codes and fine-grained complexity.

Education

Rutgers University
PhD in Computer Science

New Brunswick, USA
2023 – present

Bangladesh University of Engineering and Technology
BSc in Computer Science and Engineering

Dhaka, Bangladesh
2016 – 2021

Publications

PREPRINTS

Ulam Rank Aggregation Is Hard to Approximate for Four Rankings

with Sk Ruhul Azgor, Diptarka Chakraborty, Le Van Cuong, Debarati Das and Tien Long Nguyen

PUBLISHED PAPERS

An Optimal Algorithm for Binary Closest String

with Nick Fischer

Proceedings of the SIAM Symposium on Simplicity in Algorithms (SOSA), 2027.

Constant Rate Isometric Embeddings of Hamming Metric into Edit Metric

with Sudatta Bhattacharya, Sanjana Dey, Elazar Goldenberg, Bernhard Haeupler, Karthik C. S. and Michal Koucký

Proceedings of the International Colloquium on Automata, Languages, and Programming (ICALP), 2026.

Algorithmic Improvements to List Decoding of Folded Reed–Solomon Codes

with Vikrant Ashvinkumar and Shashank Srivastava

Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (SODA), 2026.

Hardness of Median and Center in the Ulam Metric

with Nick Fischer, Elazar Goldenberg and Karthik C. S.

Proceedings of the European Symposium on Algorithms (ESA), 2025.

Explicit Good Codes Approaching Distance 1 in Ulam Metric

with Elazar Goldenberg and Karthik C. S.

Proceedings of the IEEE International Symposium on Information Theory (ISIT), 2024.

IEEE Transactions on Information Theory, Volume 71, Issue 7, July 2025.

Selected Talks

Algorithmic Improvements to List Decoding of Folded Reed–Solomon Codes

2026 ACM-SIAM Symposium on Discrete Algorithms, Vancouver, BC, Canada
NYC Student Theory Day, New York, NY, USA

January 2026
November 2025

Hardness of Center and Median in the Ulam Metric

2025 European Symposium on Algorithms, Warsaw, Poland

September 2025

Constant Rate Isometric Embeddings of Hamming Metric into Edit Metric

Columbia Theory Lunch, New York, NY, USA
NYU Theory Seminar, New York, NY, USA
Rutgers/DIMACS Theory of Computing Seminar, Piscataway, NJ, USA

April 2025
April 2025
January 2025

Explicit Good Codes Approaching Distance 1 in Ulam Metric

2024 IEEE International Symposium on Information Theory, Athens, Greece
DIMACS Workshop on Efficient Algorithms for High Dimensional Metrics

July 2024
May 2024

Academic Honors & Awards

- 2026 Rizvi Research Excellence Award, Rutgers Computer Science Department
- 2026 Research and Travel Award, Rutgers School of Graduate Studies
- 2024, 2025 Outstanding Teaching Assistant Award, Rutgers Computer Science Department
- 2024 ISIT 2024 Student Travel Grant, IEEE Information Theory Society
- 2017–2021 University Merit Scholarship, Bangladesh University of Engineering and Technology
- 2017–2021 Dean’s List Award, Bangladesh University of Engineering and Technology
- 2015 Champion, Bangladesh Mathematical Olympiad

Service

SERVED AS REFEREE

Conferences: SOSA 2027, FSTTCS 2026, RANDOM 2026, MFCS 2026, ESA 2026, ICALP 2026, ITCS 2026, SOSA 2026, CCC 2025, ESA 2024, WALCOM 2023, WALCOM 2021, COCOON 2021, IWOCOA 2020.

Journals: Theoretical Computer Science, Algorithmica, Journal of Graph Algorithms and Applications.

OTHER SERVICES

- 2026 Co-organizer, Rutgers/DIMACS Theory of Computing Seminar
- 2022 Deputy Leader, Team Bangladesh, European Girls’ Mathematical Olympiad
- 2021 Observer, Team Bangladesh, International Mathematical Olympiad
- 2020–2021 Academic Coordinator, Bangladesh Mathematical Olympiad

Research Visits

University of Illinois Urbana-Champaign
Host: Fernando Granha Jeronimo

Illinois, USA
Summer 2026

Institute for Computer Science, Artificial Intelligence and Technology (INSAIT)
Host: Amir Abboud and Bernhard Haeupler

Sofia, Bulgaria
Fall 2024

Teaching Experience

AT RUTGERS

- Spring 2025 Automata and Formal Languages (01:198:452), Teaching Assistant
- Spring 2024 Discrete Structures I (01:198:205), Teaching Assistant
- Fall 2023 Automata and Formal Languages (01:198:452), Teaching Assistant

OTHER TEACHING

- July 2024 DIMACS Tutorial on Fine-grained Complexity, Teaching Assistant